J-C-103e
February 19, 1963
SUPERSEDING
Int. Fed. Spec. J-C-00103d(GSA-FSS)
June 20, 1962 and
Fed. Spec. J-C-103b
May 2, 1960

FEDERAL SPECIFICATION

CABLE, POWER, ELECTRICAL (RUBBER- INSULATED, GENERAL PURPOSE) AND WIRE, ELECTRICAL (RUBBER-INSULATED, GENERAL PURPOSE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cable and wire which employ copper or aluminum, solid or stranded conductors, insulated with rubber (see 6.4) for 0-to 5000-volt service.

1.2 Classification.

1.2.1 *Types.* Cable and wire furnished under this specification shall be of the following types, as specified, (see 6.1).

Type designation	Insulation	Maximum working volts	Temperature limit	
			Dry	Wet
RHW	*Heat and moisture resistant	5000	75	75
RHH	*Heat resistant	600	90	—
SA	*Silicone heat resistant ...	600	90	—

* Rubber (see 6.4)

1.2.1.1 The following letter suffixes indicate modifications of the type designations in 1.2.1 as specified in the invitation for bids, (see 6.1):

(D) designates a twin wire or cable having two conductors laid parallel under an outer covering.

(M) designates a cable having two or more conductors twisted together under an outer covering.

(L) designates a wire or cable having a lead sheath.

(AL) designates a wire or cable having an aluminum sheath.

(B) designates a wire or cable having a braided jacket.

(N) designates a wire or cable having a neoprene jacket.

For example, the combination of letters RHH-ND represents a twin wire cable having two conductors each covered with heat resistant insulation and a neoprene jacket laid parallel under an outer neoprene jacket, and RHH-B represents a single conductor having heat resistant insulation and a braided jacket.

1.2.2 *Classes.* Cable and wire furnished under this specification shall be of the following stranding classes as specified.

Class 1.—Solid conductor (AWG 14-AWG 8 only).

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Class 2. — Concentric-lay-stranded conductors of standard flexibility, conforming to class B of ASA. Standard C7.8 or C7.21.

Class 3.—Concentric-lay-stranded conductors when greater flexibility is desired, conforming to class C of ASA. Standard C7.8 or C7.21.

Class 4.—Rope-lay-stranded conductors when extreme flexibility is desired, conforming to class G of ASA. Standard C7.13.

1.2.3 *Kinds.* Cable and wire furnished under this specification shall be of the following kinds, as specified.

Kind A.—Aluminum conductor(s) (types RHH and RHW only).

Kind C.—Copper conductor(s).

1.2.4 *Voltages.* The maximum working voltages of wire and cable in accordance with this specification shall be 600, 1,000, 2,000, 3,000, or 5,000 volts, as specified in the invitation for bids. The letter designations shown in 1.2.1 when used alone indicate conductors for use at not more than 600 volts. Cable and wire for use at higher voltages shall be indicated by adding numeral suffixes to the type letters in 1.2.1 and 1.2.1.1 as follows:

10 for use at not more than 1,000 volts.

20 for use at not more than 2,000 volts.

30 for use at not more than 3,000 volts.

50 for use at not more than 5,000 volts.

For example, RHH-N represents a single conductor having heat resistant insulation and a neoprene jacket designed for use at not more than 600 volts and RHW-N20 represents a single conductor having heat resistant insulation and a neoprene jacket designed for use at not more than 2000 volts.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

J-C-98—Cable and Wire, Insulated; Methods of Sampling and Testing.

UU-P-271—Paper, Wrapping, Waterproofed, Kraft.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547—Liner, Case, Waterproof.

MIL-C-12000—Cable, Cord, and Wire, Electric, Packaging and Packing.

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Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109—Inspection Terms and Definitions.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Underwriters' Laboratories, Inc., Publication:

UL-44—Standard for Rubber-Covered Wires and Cables.

(Copies may be obtained from the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, N. Y.; 207 East Ohio Street, Chicago 11, Illinois; or 1655 Scott Blvd., Santa Clara, California.)

American Standards Association Publications:

C1—National Electrical Code.
C7.8—Concentric-Lay-Stranded Copper Conductors, Hard-Medium-Hard, or Soft, Specification for.
C7.13—Rope-Lay-Stranded Copper Conductors Having Concentric-Stranded Members, for Electrical Conductors, Specifications For.
C7.21—Concentric-Lay-Stranded Aluminum Conductors, Hard-Drawn, Three-Quarter Hard-Drawn, And One-Half Hard-Drawn, Specification for.

(Copies may be obtained from American Standards Association, Inc., 10 East 40th Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Cable and wire furnished under this specification shall conform to the requirements of Underwriters' Laboratories, Inc., Standard UL-44 for Rubber-Covered Wires and Cables, and the pertinent American Standards Association, Inc., Standards (1.2.2).

3.1.1 *Conductor size.* Conductors in cable and wire furnished under this specification, shall range in size from AWG14 through 2,000,000 circular mil, as specified (6.1).

3.2 For a voltage rating greater than 2000 volts, type RHW insulation is required to be ozone resistant, and suitable shielding is required in accordance with ASA. Standard C1.

3.3 Fire and casualty hazards.

3.3.1 Each bidder shall submit to the contracting agency proof that the wire and cable he proposes to supply under this specification conforms to the requirements of Underwriters' Laboratories, Inc., Standard UL-44 for Rubber-Covered Wires and Cables. The label or reexamination listing of the Underwriters' Laboratories, Inc., may be accepted as evidence that the wire or cable conforms to this requirement.

3.3.2 In lieu of the label or reexamination listing, the bidder may submit independent proof satisfactory to the contracting agency that his cable or wire conforms to the applicable requirements of the published standards, including methods of tests, of the Underwriters' Laboratories, Inc., Standard UL-44 for Rubber-Covered Wires and Cables.

3.3.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his material.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the perform-

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ance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Unless otherwise specified, inspections by the supplier may be performed on a statistical or production lot basis to assure that the quality of manufacture is equal to the standards set by Underwriters' Laboratories, Inc. (6.2).

4.2 Sampling, nonmilitary.

4.2.1 *Lot.* Unless otherwise specified in the invitation for bids, the lot shall be formed as described in paragraph 3 of section 6 of J-C-98.

4.2.2 *Sampling for inspection and tests.* Unless otherwise specified in the invitation for bids, the sample shall be taken as described in paragraph 6 of section 6 of J-C-98.

4.3 Visual inspection, nonmilitary.

4.3.1 Each inspection unit in the sample, 4.2.2, as well as the package from which the inspection unit is taken, shall be examined visually before any tests are made. Visual inspection shall cover construction, tinning of conductors, splices, marking, type of conductor, lubrication of neoprene jacket, joints and repairs, and may be limited to that part of the package which is visible without unrolling.

4.4 General inspection, military.

4.4.1 *Definitions.* MIL-STD-109 applies for definitions of inspection terms used herein.

4.4.2 *Acceptance inspection.* Acceptance inspection shall consist of group A, group B, and group C inspection in 4.4.2.1 through 4.4.2.3.1. Group B inspection shall normally be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group B or group C inspection or both.

4.4.2.1 *Group A inspection.* Group A inspection including sampling shall be in accordance with table I and MIL-STD-105 for ordinary inspection. Group A inspection shall be performed in any order which is satisfactory to the Government inspector.

TABLE I.—*Group A inspection*

Inspection	AQL	
	Major	Minor
	Percent	*Percent*
Visual and mechanical:		
General construction		
Tinning		
Splices		
Marking		
Type of conductor (diameter)	1.0	4.0
Lubrication of jacket		
Joints		
Repairs		
Electrical:		
Resistance of conductor		
Insulation resistance	1.0	*
Dielectric strength		

* All electrical defects are considered major.

4.4.2.2 *Group B inspection.* This inspection, including sampling, shall conform to table II and to MIL-STD-105 for small sample inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1.

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4.4.2.2.1 *Group B sampling plans.* The group B sampling plans as listed in table II shall be as follows:

AQL	Inspection level for normal and tightened inspection	Inspection level for reduced inspection
Percent		
6.5	L-7	L-5

TABLE II.—*Group B inspection*

Inspection	Sampling plan
Subgroup 1:	
Average thickness of insulation	See 4.4.2.2.1
Minimum thickness of insulation (less than nominal 3/64 inch)	
Minimum thickness of insulation (nominal 3/64 inch and over)	
Thickness of braid	
Thickness of cotton tape	
Thickness of lead covering	
Thickness of neoprene jacket average and minimum	
Lay of pitch, lay of twist	
Tangent of angle between braid and axis of wire or cable	
Tangent of angle between serving wrap and axis of wire or cable	
Picks per inch	
Yarn size	
Length of lay, wrap or serving	
Carriers and ends per carrier, braids	
Subgroup 2:	
Accelerated aging	6 specimens per 100,000 feet
Air oven test	
Oxygen pressure test	
Air pressure test	
Change in tensile strength after oil immersion	
Subgroup 3:	6 specimens per 100,000 feet or fraction thereof
Flame resistance, fibrous covering except tapes	
Flame resistance, neoprene jacket	
Subgroup 4:	
Moisture resistance, fibrous covering other than tape	6 specimens per 100,000 feet or fraction thereof
Moisture absorption, insulation	

4.4.2.2.2 *Order of inspection within group B.* Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

4.4.2.2.3 *Procedure in the case of failure.* If an inspection lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.4.2.3 *Group C inspection.* This inspection shall be as listed in table III.

TABLE III.—*Group C inspection*

	Sampling plan
Recovery—	
Tensile strength, insulation and neoprene jacket.	6 specimens per 500,000 feet
Elongation, insulation and neoprene jacket	
Specific inductive capacity	

4.4.2.3.1 *Specimens.* Six specimens shall be prepared and tested from each lot of 500,000 feet or fraction thereof of cable and wire. If each of the 6 specimens meet the specified requirements, the lot shall be accepted.

4.4.2.3.1.1 If 2 or more of the 6 specimens fail to meet the specified requirement, the lot shall be rejected.

4.4.2.3.1.2 If only 1 of the 6 specimens fails to meet the specified requirements, a second sample of 6 specimens shall be tested. Lot acceptance shall be permitted only if in the combined sample of 12 specimens not more than one specimen fails to meet the requirements. Failure of 2 or more specimens in the combined first and second samples shall result in lot rejection.

4.4.2.3.2 *Noncompliance.* If a lot fails group C inspection, the contractor shall immediately investigate the cause of failure

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and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and on all units of product which were manufactured with the same conditions, materials, processing, etc. If the Government inspector does not consider that the corrective action will enable the product to meet the special requirement or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.3 *Disposition of nonconforming product.* Disposition of lots and sample units rejected during group A or B inspection shall be in accordance with MIL-STD-105. The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means to correct the fault:

(*a*) Sample units found defective during groups A, B, or C inspection.
(*b*) Inspection lots rejected during group A or B inspection.
(*c*) Product has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.5 Tests.

4.5.1 *Nonmilitary.* The cable and wire shall be subjected to the tests set forth in Underwriters' Laboratories, Inc. Standard UL-44 to determine compliance with 3.1.

4.5.2 *Military.* The cable and wire shall be subject to the tests set forth in Underwriters' Laboratories, Inc. Standard UL-44, modified by the instructions in 4.4 to determine compliance with 3.1.

4.6 Compliance criteria, nonmilitary.

4.6.1 *Inspection.* If one or more of the packages 4.2.2, fails to conform to the requirements for which inspection was made, 4.3, the lot from which the packages were taken shall be considered to have failed to meet the requirements of this specification.

4.6.2 *Tests.* Compliance of the lot with respect to the tests shall be based on the principles described in section 6 of J-C-98.

4.6.2.1 Any reel, coil, or roll of cable or wire which shows the presence of a defect or fails to meet the requirement when subjected to the electrical tests, shall be considered to have failed to meet the requirements of this specification. If after the defects are repaired and the roll, reel, or coil is again subjected to the dielectric strength and insulation resistance tests, the presence of a defect is not shown, the roll, reel, or coil shall be considered to have met the requirements of this specification for dielectric strength and insulation resistance.

4.7 Compliance criteria, military.

4.7.1 The compliance criteria shall be as set forth in 4.3.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the various levels of packaging and packing shall be as specified in Fed. Std. No. 102.)

5.1 Packaging.

5.1.1 *Civil agencies.*

5.1.1.1 *Levels A. and B.*

5.1.1.1.1 *Coils (AWG Sizes 14 to AWG 10).* Cable and wire shall be uniformly coiled and packaged 500 feet in a fiberboard box conforming to PPP-B-636, type I or type II, class 2, provided with the special knock-out feature normally used in commercial practice for dispensing purposes.

5.1.1.1.2 *Coils (AWG Sizes 8 to AWG 2).* Cable and wire shall be uniformly coiled and packaged 500 feet per coil. The coils shall be spirally wrapped with waterproofed kraft

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paper conforming to UU-P-271, class E-2 and sealed.

5.1.1.1.3 *Reels.* Cable and wire shall be packaged on non-returnable wood or metal spools or reels suitably designed with strong flanges of sufficient diameter to protect the cable or wire from damage during shipment and storage. After the cable or wire has been wound on the reel, it shall be covered with a full wrap of waterproof kraft barrier material conforming to UU-P-271, class E-2. The wrap shall extend between the flanges and be secured in place with a minimum of two flat steel straps.

5.1.1.2 *Level C.* Cable and wire shall be packaged in coils or on reels of a kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.1.2 *Military agencies.*

5.1.2.1 *Level A.* The cable and wire shall be packaged in accordance with Level A of MIL-C-12000.

5.2 Packing.

5.2.1 *Civil agencies.*

5.2.1.1 *Level A.*

5.2.1.1.1 *Coils, (AWG sizes 14 to AWG 10).* Cable and wire packaged as specified in 5.1.1.1 shall be packed in nailed wood or wood cleated plywood boxes conforming to PPP-B-621 and PPP-B-601, respectively. Containers shall be fitted with waterproof case-liners conforming to MIL-L-10547, class 2, and sealed in accordance with the appendix thereto.

5.2.1.1.2 *Coils (AWG sizes 8 to AWG 2), and reels.* Cable and wire packaged as specified in 5.1.1.1.2 and 5.1.1.1.3, shall require no further overpacking.

5.2.1.2 *Level B, coils and reels.* Cable and wire packaged as specified in 5.1.1.1.1, 5.1.1.1.2 and 5.1.1.1.3 shall require no further overpacking.

5.2.1.3 *Level C.* The subject commodity shall be packaged and packed in substantial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.2 *Military agencies.*

5.2.2.1 *Level A or B.* The cable and wire shall be packed in accordance with level A or B, as specified in MIL-C-12000.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.
(*b*) Length or unit of issue required.
(*c*) Type (1.2.1, 1.2.1.1), class (1.2.2), kind (1.2.3), and service voltage (1.2.4) of cable and wire.
(*d*) Size of conductor (3.1.1).
(*e*) Administrative provisions for inspection records (4.1).
(*f*) Basis of inspection, if other than specified (4.1.1).
(*g*) Lot, if other than specified (4.2.1).
(*h*) Sample, if other than specified (4.2.2).

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(*h*) Level of packaging and packing required (5.1 and 5.2).
(*i*) Special marking, if required (5.3).

6.2 If laboratory tests by the government are required, purchasing officers should order cable or wire for test purposes in addition to the number of feet required for installation.

6.3 Underwriters' Laboratories, Inc. Standard UL-44 for Rubber-Covered Wire and Cable permits a substitution of the spark test in place of part of dielectric strength and insulation resistance tests for 600-volt wire and cable. Since the spark test is usually conducted as a part of the manufacturing process, the dielectric strength and insulation resistance (when required) should be specified in the invitation for bids and arrangements should be made with the manufacturer for witnessing the tests in his plant.

6.4 Throughout the specification the term "rubber" is used to designate compounds employing natural, SBR, or butyl synthetic rubber, or blends thereof; also silicone rubber. The term "neoprene" is used to designate the synthetic rubber "polychloroprene".

6.5 Class RHW fibrous-covered wire is recognized by Underwriters' Laboratories, Inc., for rating greater than 2000 volts only if provided with suitable shielding.

6.6 **Transportation description.** Transportation description and minimum weights applicable to this commodity are:

Rail—Copper
Cable, Electric, Copper, Not Otherwise Indexed by Name.
Carload minimum weight 30,000 Pounds.

Motor—Copper
Cable, Electric, Copper, Not Otherwise Indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Rail—Aluminum
Cable, Electric, Aluminum.
Carload minimum weight 30,000 pounds.

Motor—Aluminum
Cable, Electric, Aluminum.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Rail—Aluminum
Wire, Aluminum.
Carload minimum weight 30,000 pounds.

Motor—Aluminum
Wire, Aluminum.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Rail—Copper
Wire, Copper, Insulated.
Carload minimum weight 30,000 pounds.

Motor—Copper
Wire, Copper, Insulated.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

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MILITARY INTEREST:
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Copies of this specification may be purchased for ¯ cents each.

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